Title: Prediction of Students' Performance Using Multiple Linear Regression Model and Random Forest Model

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CHAPTER 1
INTRODUCTION

- **1.1:Background of the study**
- Student academic performance reflects the effectiveness of teaching and learning, but traditional assessment methods are reactive and limit early intervention.
- □ The rise of digital learning systems has generated large educational datasets (grades, attendance, engagement), enabling deeper analysis of student outcomes.
- □ Educational Data Mining and Learning Analytics support proactive, data-driven decision-making in education.
- □ Machine learning models, including Linear Regression and Random Forest, can predict performance and identify at-risk students.
- □ This study aims to improve prediction accuracy and support evidence-based interventions in education.

Problem Statement

- Many institutions still use reactive methods, identifying student performance issues only after exams, limiting early intervention.
- Although large student datasets exist, predictive analytics is underused, especially in manual or fragmented systems.
- Most studies use a single model, limiting comparison of accuracy and interpretability.
- This study compares Linear Regression and Random Forest to improve prediction and identify key performance factors.

Objectives of the Study

- The main objective is to analyze and predict student academic performance using Linear Regression and Random Forest models.
- **Specific Objectives**
- □ To fit a Linear Regression model for predicting academic performance.
  □ To fit a Random Forest model for prediction.
  □ To perform diagnostic tests for model evaluation.
  □ To compare the accuracy and interpretability of both models.

Scope of the study

- This study used a Kaggle student dataset containing academic records such as exam scores, assignments, attendance, demographics, and engagement indicators.
- □ The study focused only on quantitative variables, excluding qualitative factors like motivation, attitudes, and learning environment due to data limitations.
- □ The analysis applied Linear Regression and Random Forest to model both linear and non-linear relationships in predicting academic performance.
- □ The dependent variable was final academic performance (grades/marks), with emphasis on predictive accuracy and model comparison rather than causal analysis or real-time intervention.

Significance of the study

For **students**, early prediction of performance helps identify academic challenges and improve study strategies.

 For **educators**, predictive insights support better teaching strategies, identification of learning gaps, and targeted interventions.

 For **institutions**, data-driven predictions improve student retention, academic planning, and quality assurance.

 For **researchers**, the study contributes to literature on **educational data mining** and predictive modeling in education.

CHAPTER 2:LITERATURE REVIEW

- **RELATED LITERATURE**
- □ Student performance prediction has been widely studied using statistical and machine learning methods.
  □ Models commonly used include Linear Regression, Decision Trees, Support Vector Machine, and Random Forest.
  □ Studies show that Random Forest and ensemble methods often achieve higher accuracy in prediction tasks (Schwarcz et al., 2021; Nahar et al., 2021).
  □ Academic performance is strongly influenced by attendance, engagement, and prior achievement (Kolo et al., 2020; Imran et al., 2019).
  □ However, Linear Regression remains important due to its interpretability (Ahmed & Elaraby, 2019).

LIMITATIONS & RESEARCH GAP

- □ Many studies rely on single-model approaches, limiting comparative analysis.
  □ Issues of missing data, small datasets, and poor generalizability are common (Materechera, 2020; Hussain et al., 2024).
  □ A persistent challenge is the trade-off between accuracy and interpretability in ML models.
  □ Few studies integrate both statistical (MLR) and machine learning (Random Forest) approaches in one framework.
  □ This study addresses these gaps using Kaggle data, OLS feature selection, MLR, and Random Forest models.

CHAPTER 3: METHODOLOGY

- □ Quantitative predictive modeling approach was used
  □ Aim: predict student academic performance (G3)
  □ **Models used:**
- **Multiple Linear Regression (MLR):**
- Interpretable model
- Shows relationship between predictors and outcome
- □ **Random Forest:**
- Handles non-linear relationships
- More accurate and robust predictions
- □ Both models used for comparison of accuracy vs interpretability

DATA DESCRIPTION AND PREPROCESSING

- **DATA DESCRIPTION**
- Kaggle dataset (student-por.csv)
   649 observations, 33 variables
   Includes academic, demographic, and behavioral factors
   Target variable: final grade (G3)

- **DATA PREPROCESSING**
- Data cleaning and handling of missing values
   Encoding of categorical variables

MODEL EVALUATION

- Data split: 80% training, 20% testing
   Evaluation metrics:
- R^2 (model fit)
- MAE (error)
- RMSE (prediction accuracy)
- Random Forest also evaluated using feature importance

CHAPTER FOUR: DATA ANALYSIS AND RESULTS

- **Introduction**
- Overview of dataset analysis for predicting student performance (G3)
- Use of two models:
  - Multiple Linear Regression (MLR)
  - Random Forest Regression

- **Data Inspection & Feature Engineering**
- Dataset contains 649 records, 33 variables
- No missing or duplicate values
- Feature engineering:
  - Total Alcohol Index
  - Binary Failure Status
- Encoding:
  - One-hot encoding for categorical variables
- Target variable: G3 (Final Grade)

Descriptive Statistics

Students are mostly aged around 16–17 years

Academic performance shows moderate variation (G3 std dev ≈ 3.23)

Absences show high variability, indicating inconsistent attendance patterns

Grades improve slightly from G1 → G3

Variable	Mean	Std Dev	Min	Max
Age	16.74	1.22	15	22
Study Time	1.93	0.83	1	4
Absences	3.66	4.64	0	32
G1	11.40	2.75	0	19
G2	11.57	2.91	0	19
G3	11.91	3.23	0	19

Exploratory Data Analysis (EDA)

Academic Momentum: There is a strong positive relationship between midterm grades (G1, G2) and final performance (G3), showing that early success strongly predicts final outcomes.

Negative Influences: High absenteeism and previous academic failures are consistently associated with lower final grades.

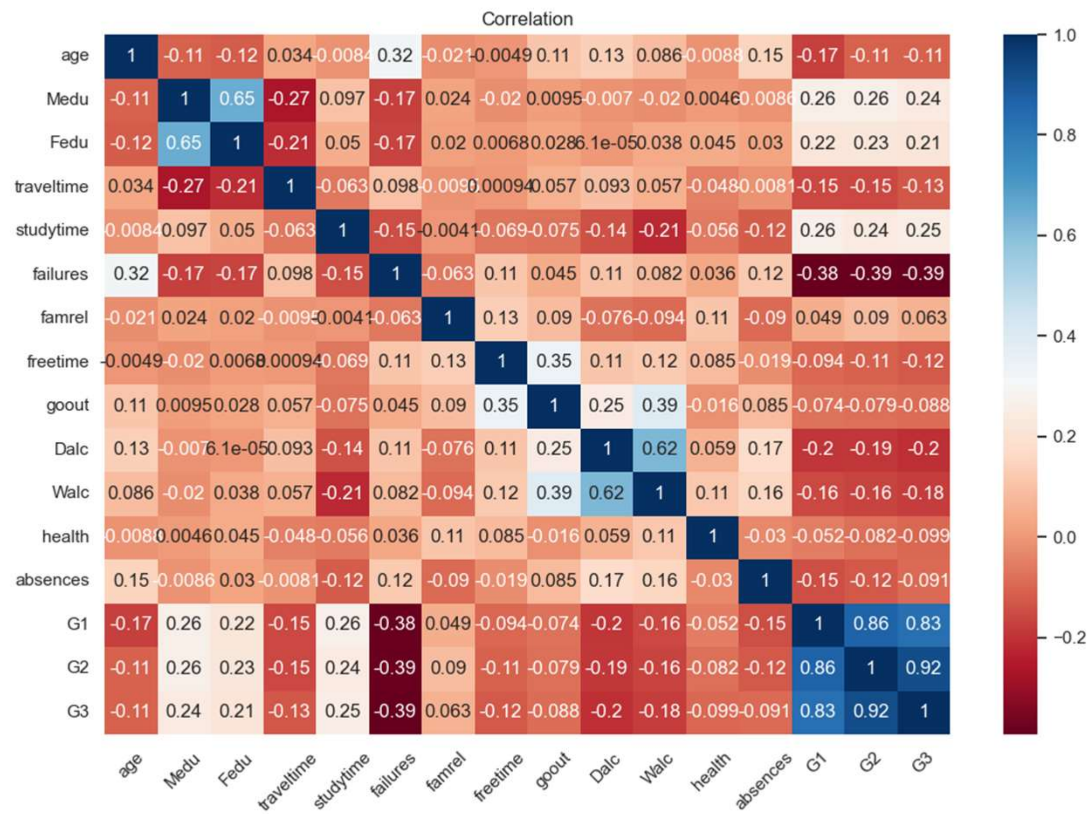
Performance Trend: Students who perform well in earlier terms are highly likely to maintain strong performance in the final exams.

Home Environment: Students with parents in professional fields (e.g., teaching and health) tend to achieve slightly higher grades.

Risk Indicators: Frequent absences and a history of failure serve as key warning signs for potential academic underperformance.

Correlation heatmap

- Pearson correlation was used to measure linear relationships between variables (-1 to +1).
- Final grade (G3) is strongly driven by G2 (0.92) and G1 (0.83), confirming a cumulative academic pattern.
- Previous failures negatively impact performance (~ -0.39 across grades).
- Parental education and study time show weak positive effects, while social factors have weak negative effects.
- Some predictors are highly correlated, indicating possible multicollinearity, while variables like health show minimal impact.

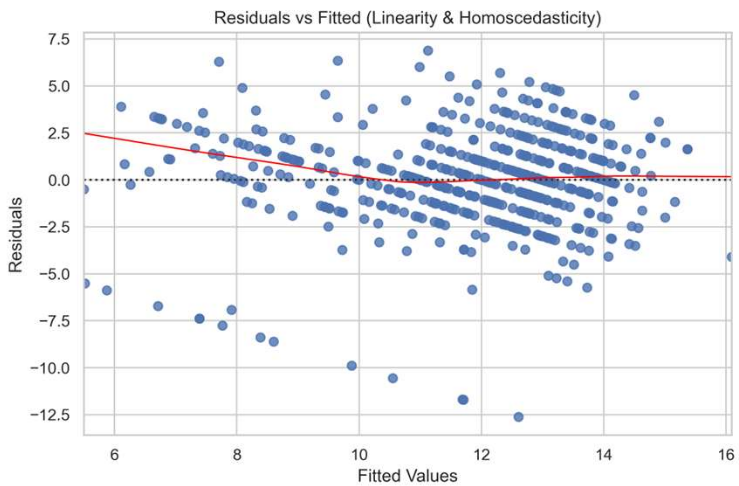


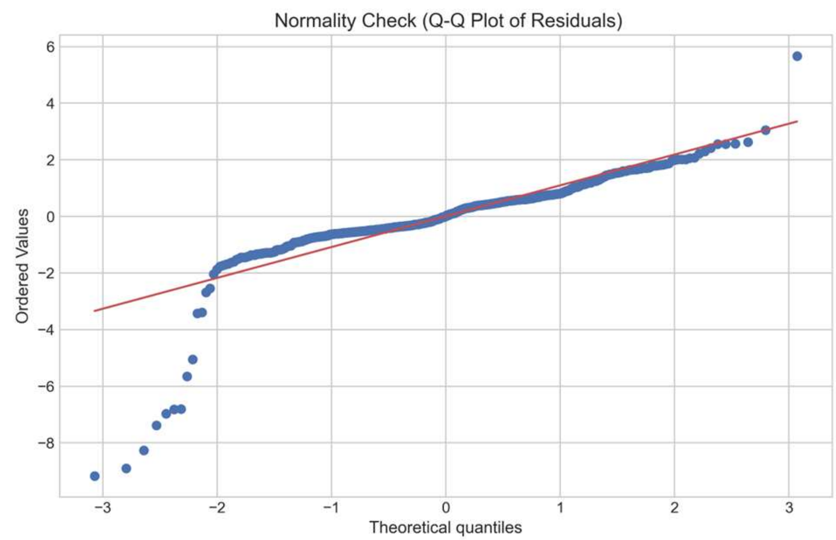
Multiple Linear Regression (MLR)

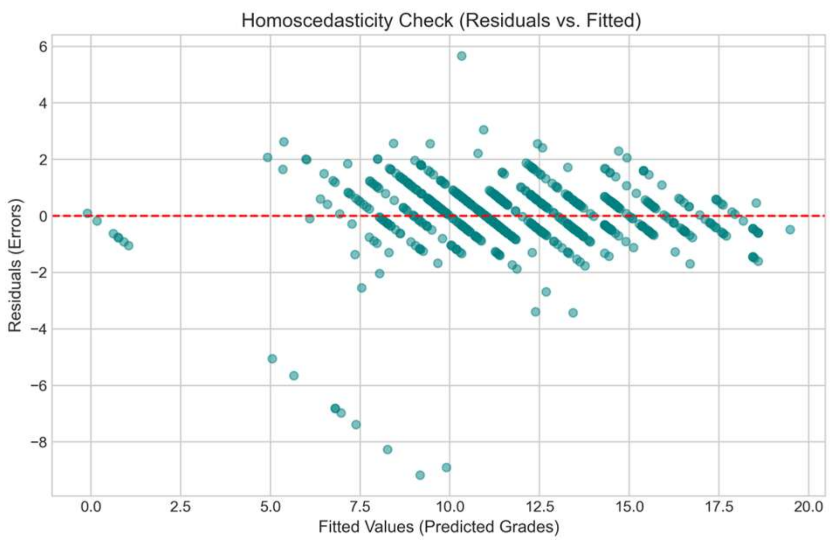
- •Strong model performance ($R^2 \approx 0.861$)
- •Key predictors:
- •G2 (strongest)
- •G1
- •Past failures
- •Most socio-demographic variables are insignificant

- **Feature Selection & Refinement**
- Backward elimination applied to variables with a p-value > 0.05.
- Final model reduced to key predictors:
  - G1, G2, absences, failure history
- Improved interpretability and stability

Diagnostic Checks



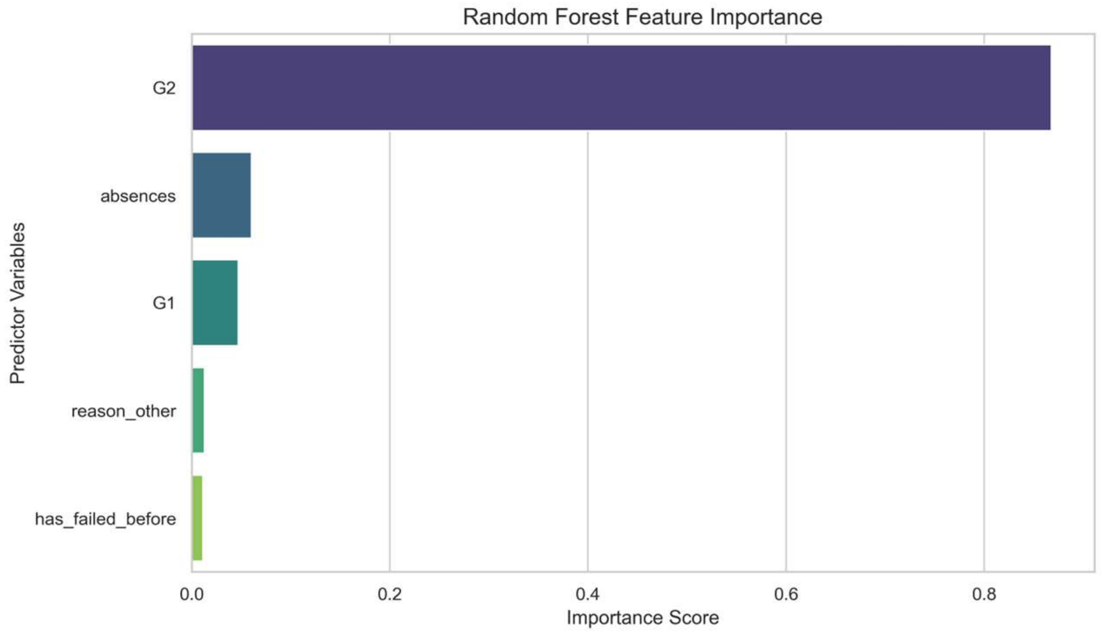




- **Diagnostic Checks**
- Linearity assumption satisfied
- Mild heteroscedasticity at extreme values
- Residuals approximately normal
- No autocorrelation (DW ≈ 1.87)
- Some multicollinearity in G1 & G2

Random Forest

- Ensemble model used for non-linear relationships
- Performance:
- $R^2 \approx 0.851$
- Key predictors:
- G2, G1, absences
- More flexible but less interpretable than MLR
- Feature importance summary
- G2 is the strongest predictor, followed by absences and G1.
- Highlights the role of academic momentum and attendance.
- Random Forest captures nonlinear effects, though MLR is slightly more accurate.



Model Comparison & Conclusion

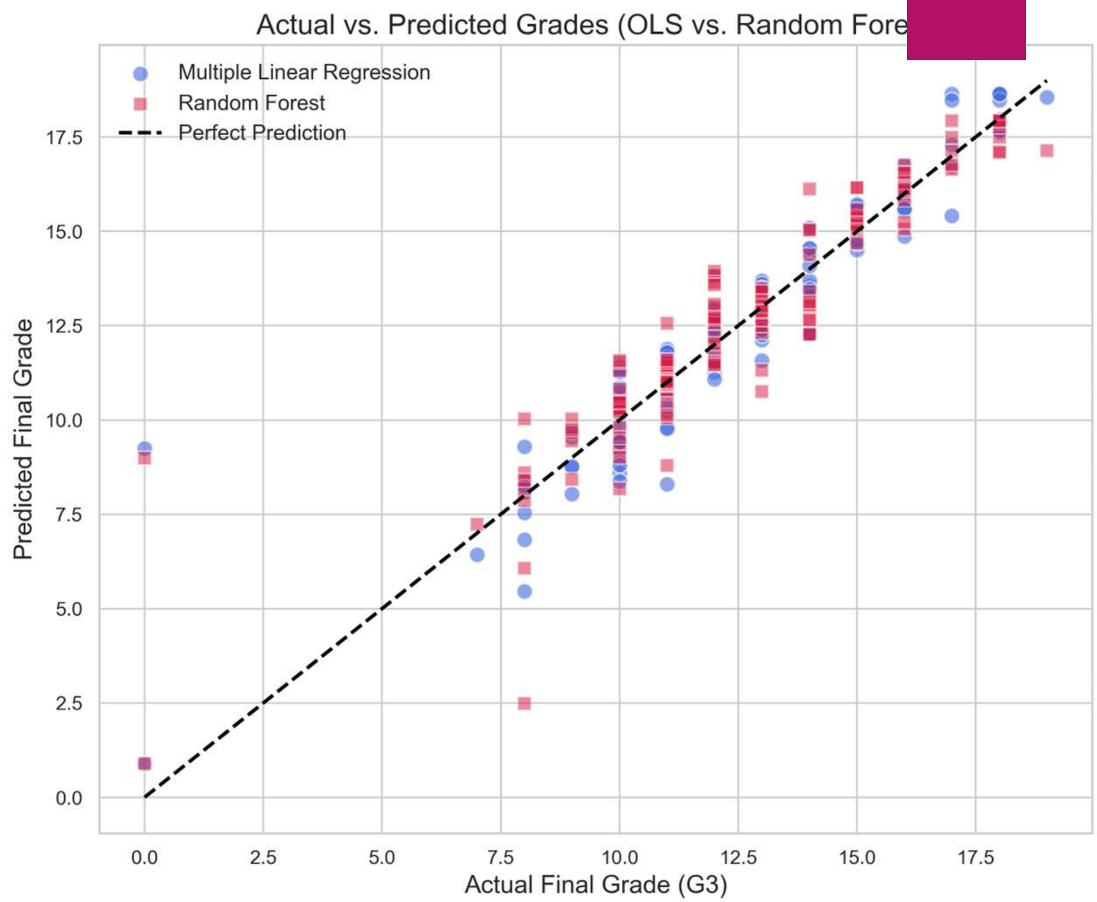
Model Performance Comparison (20-Point Scale)

- **Accuracy Winner: Multiple Linear Regression (MLR)**
  - **MAE:** 0.7180 (Avg. error of **0.72 points**)
  - **RMSE:** 1.1403 (Standard error of **1.14 points**)
  - **Result:** MLR achieved a lower error margin than Random Forest (**0.75** MAE).
- **Predictive Power (R^2)**
  - **MLR: 86.67%** of variance explained.
  - **RF: 85.09%** of variance explained.
  - **Result:** MLR captured the data trends more effectively.
- **Precision in Context**
  - The total percentage error was only **3.6%**.
  - Strong linear correlations between G1, G2, and final G3 grades favored the MLR approach.
- **Key Conclusion**
  - While Random Forest is robust, **MLR provided higher precision and better interpretability** for simulating student outcomes.

Metric	MLR (Winner)	Random Forest
R-Squared	**0.8667**	0.8509
MAE	**0.7180**	0.7551
RMSE	**1.1403**	1.2600

Performance Visualization

- **Observations:** MLR (Blue) clusters tighter to the "Line of Perfection."
- Both models struggle with "Zero" outliers.
- High density near the diagonal confirms robust prediction for the most students.



CHAPTER 5: CONCLUSION

The project applied Multiple Linear Regression and Random Forest to predict student academic performance.

Multiple Linear Regression achieved better performance ($R^2 = 0.8667$, lower errors), showing a stronger fit.

v This confirms that the relationship between academic variables is largely linear, especially the impact of G1 and G2 on G3.
v The findings highlight that academic momentum, attendance, and past failures are the key drivers of performance.
v Overall, the study contributes to educational analytics by improving performance prediction and supporting early intervention strategies.

Recommendations

Implement automated Early Warning Systems (EWS) using predictive models based on early academic performance to identify at-risk students and enable timely interventions such as tutoring and counseling.

Strengthen attendance monitoring by setting clear absence thresholds that trigger immediate intervention, as absenteeism is a strong predictor of poor performance and may reflect underlying social, health, or motivational issues.

Provide targeted support programs for students with a history of academic difficulties through remedial classes and psychological support at the start of the academic year.

Incorporate richer behavioral data such as digital learning engagement and library usage to improve model accuracy and provide a more comprehensive understanding of student performance factors.